Improve One Cause

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Teach repair as controlled evidence, including a no-change result.

Model S0/S1 and the T1 regression, keep actual traces separate and reserve the independent equipment repair.

Keep one habit: Look for the exact changed instruction, preserved requirements and complete regression evidence.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: **Repair** a located **defect** without changing the agreed requirements

Repair and **regression** testing concern the implementation of explicit rules, not **fitting** a model to examples. S0 and T1 are deliberately faulty instructional examples.

Pupils whose P1 works should not invent a **defect** or claim an unneeded improvement. Assess the causal explanation and retained **evidence**, not only a favourable count.

Final evaluation tests a fixed version; development success and a polished presentation cannot replace it.

By the end, I can **repair** one identified cause and rerun cases that previously passed.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Saved P1 and the clearly labelled S0, S1 and T1 comparison instructions described in the lesson
- Catalogue B versions 1, 2 and 3, the manual baseline and all earlier development records
- Blank **change record**s and an envelope or folder for the candidate; keep reserved final cases teacher-only

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Test Edge Cases and Fairness. Give an example and a mistake that the idea helps you avoid.

Look for: Choose tests just below, at and above a decision boundary

Reference example, not the only acceptable wording: 9 Yes has no match: B1 and B3 are too long and B2 is not quiet; B4 also fails. 10 Yes and 11 Yes select B1, which is quiet and fits at equality or with time left.

Fixing the other conditions isolates the time comparison. S rejects equality at B1 and cannot find a later fit for 10 Yes, so incorrectly returns no match.

Without opening the old lesson, explain one key idea from Build a Paper Prototype. Give an example and a mistake that the idea helps you avoid.

Look for: Build a runnable paper prototype with explicit instructions and visible state

Reference example, not the only acceptable wording: The kit contains labelled instructions and catalogue, request fields, a marker and a trace sheet with run and version labels.

The request and catalogue stay fixed during the run; the current row and trace advance. The final **output** starts clear.

The cards validate, check both conditions, return the first fit, advance after failure and stop with no match after exhaustion.

Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

Look for: Describe a specific user difficulty using supplied **evidence** without inventing its cause

Reference example, not the only acceptable wording: For example: fictional pupils choosing activities in the supplied classroom exercise had to choose again in four of ten requests because the first choice did not fit the time or room conditions.

Build an AI app names a solution or construction task without specifying a user difficulty or **evidence** of need.

Time fits when required minutes are no more than available minutes, including equality. Strictly less than wrongly excludes an exact fit.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Test Edge Cases and Fairness

Build a Paper Prototype

Find a Real Problem

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Version A makes four errors. Version B makes one after lighting, labels and rules all change. What can you conclude, and how would you isolate the effect of lighting?

Concept to explain

An improvement should target **evidence** from a failure, not random change.

Term	Meaning
root cause	earliest responsible reason for a failure
controlled change	one targeted modification while other factors stay fixed
retest	repeat the same case after a change
regression	previously working behavior that becomes worse

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach a located repair while preserving the full contract

> Distinguish worked version predictions, actual learner runs and the separate U-version examples.

Read the unchanged activity catalogue and requirements

Finding a failure is useful only if you can explain and check the **repair**. This week you will locate an incorrect comparison, change it in a labelled version, and test both the revealing case and behaviour that worked before.

A **repair** can solve one problem while introducing another, so one newly passing example is not enough.

Keep catalogue B version 1 in order: B1 Bookmark drawing, 10 minutes, quiet Yes; B2 Story circle, 8 minutes, quiet No; B3 Pattern puzzle, **12 minutes**, quiet Yes; B4 Tower challenge, 15 minutes, quiet No.

These fictional values assume materials are available. **Input**s are whole minutes from 0 to 30 inclusive and Quiet required Yes or No. No allows either quiet or nonquiet activities.

A candidate fits when its duration is no more than the available minutes AND either quiet is not required or the activity is quiet. Return the first fit with a correct reason.

Valid **input** with no fit gives no match after all rows fail. Invalid or missing **input** needs clarification before search. Each new run starts with validation and a cleared **output**, then the first row if valid.

Suggestions do not book or publish anything. These **written rule**s are not a model fitted from examples.

Locate S0’s defect and specify the S1-only change

S version 0 follows the P1 **process** except that its time check uses strictly less than. On 10 Yes with catalogue B version 1, it reports no match. The expected result is B1 because 10 fits exactly and B1 is quiet.

Compare the saved instruction with the trace: the cause is the exclusion of equality, not a need to change the catalogue or expected answer.

Create S version 1 by replacing only the time comparison with no more than. Keep validation, the quiet condition, catalogue order, advancement and stopping rules.

Write a **change record** naming S0, S1, the failing 10 Yes case, the comparison changed and why equality belongs in the rule. Retain the old version and failed trace; do not overwrite them with the new result.

Expose T1’s quiet regression and plan reruns

Consider a different proposed **repair**, T version 1. It accepts equal times but accidentally removes the quiet check. It fixes 10 Yes, yet now returns B2 for 9 Yes.

That second request correctly returned no match under S0 because B2 is not quiet. Previously correct behaviour has become wrong: this is a **regression**.

Case with catalogue B version 1	Required result	Outputs from the stated instructions
10 Yes; equality repair	B1 with a valid reason	S0 no match; T1 B1; S1 B1
9 Yes; quiet regression check	No match with a valid reason	S0 no match; T1 B2; S1 no match
8 No; another equality case	B2 with a valid reason	S0 no match; T1 B2; S1 B2
6 No; exhaustion check	No match with a valid reason	S0 no match; T1 no match; S1 no match

These **output**s are worked **prediction**s from the described versions. Your own test record must contain the actual runs and their reasons. T1 should not replace the candidate merely because it passes the first row.

For S1, check the revealing equality cases and rerun all four routine cases from Week 33, then validation and the labelled catalogue variations from Week 34.

A full pass still needs both the required **output** and a correct explanation.

Introduce the separate two-activity U example

A fictional quiet-activity chooser has a boundary bug and a proposed **repair**.

A: quiet Yes, 8 minutes; B: quiet No, 5 minutes. Request quiet required Yes.

U0 accepts only duration<available time. U1 fixes ≤ but accidentally removes the quiet requirement.

Inspect the labelled repair comparison

AI ACADEMY / AI-1/35 · 03 · INSPECT THE LESSON MODEL

A repair with evidence

Locate	Change	Check
Keep the failing trace	Save a new version	Rerun the revealing case
Name the broken requirement	Correct the faulty step	Rerun earlier passing cases
Keep input and catalogue labels	Record what changed	Keep every result and limit

A passing repair test checks the reported defect. Regression tests examine previously correct behaviour.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the bad revision and preserved requirement

AI ACADEMY / AI-1/35 · 02 · TRACE THE CASE

Follow the deciding steps

01 Boundary failure

At 8 minutes, U0 rejects A and rejects noisy B: none.

02 Expected repair

Correct logic must allow A when time equals 8.

03 Regression case

At 6 quiet-required minutes, correct output is none.

04 Bad revision

U1 selects noisy B at 6; fixing equality introduced a new failure.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Boundary failure	At 8 minutes, U0 rejects A and rejects noisy B: none.
2. Expected repair	Correct logic must allow A when time equals 8.
3. Regression case	At 6 quiet-required minutes, correct output is none.
4. Bad revision	U1 selects noisy B at 6; fixing equality introduced a new failure.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **repair** must preserve the full contract, not just make one failing example pass. The quiet condition and duration comparison are separate requirements.

Changing only the inequality targets the recorded cause; removing the quiet check changes another behavior. Save previously passing cases to reveal such **regression**s.

Limit generalisation from two examples

Two examples establish only these outcomes; further cases are needed before generalising the **repair**.

Predict the targeted U2 result

Change: Use U2, which changes < to ≤ and keeps the quiet check.

Prediction: At 8 return A; at 6 return none.

Reason: Equality is **repair**ed while B remains ineligible because it is noisy.

Use the picture to identify the regression

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

> **1. Input — information entering**
>
> **INPUT** failure record and suspected cause

> **2. Process — the deciding step**
>
> **PROCESS** change one responsible rule, **input**, or threshold

> **3. Output — the result**
>
> **OUTPUT** same tests rerun with before/after **evidence**

CORE IDEA An improvement should target **evidence** from a failure, not random change.

UNDER THE HOOD Controlled iteration links an intervention to an observed outcome. Holding test cases and unrelated components constant strengthens causal inference.

Improvement claims must include **regression**s and tradeoffs, not only **repair**ed cases.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

CASE The sorter misclassifies worn labels, so the team changes the text-reading rule or adds a safe visual clue while keeping other parts fixed.

STEP 1 — OBSERVE failure record and suspected cause

STEP 2 — TRACE change one responsible rule, **input**, or threshold

STEP 3 — CONCLUDE same tests rerun with before/after **evidence**

MODEL REASONING Name the **evidence**, suspected cause, single change, and predicted effect before editing. Rerun the full fixed suite, compare case by case, and keep the change only if benefits outweigh new failures.

DO NOT OVERCLAIM A higher pass count can hide a serious new failure; review every changed outcome.

What new **evidence** could change this answer?

Concrete teaching case

The team changes lighting, labels and rules together; errors fall from four to one.

Can it identify the cause of improvement?

Exact explanation to model

No. Several factors changed. Compare one planned change at a time with controlled conditions.

Explain the added worked example

Key terms

Term	Meaning
Defect	A fault in the instructions or implementation that can cause incorrect behaviour

Term	Meaning
Repair	A change intended to correct a located defect
Regression	Previously correct behaviour that becomes incorrect after a change
Change record	A note of what changed, why, and which versions and tests support it
Fixed candidate	A saved version kept unchanged while a specified evaluation is run

Model the reasoning aloud

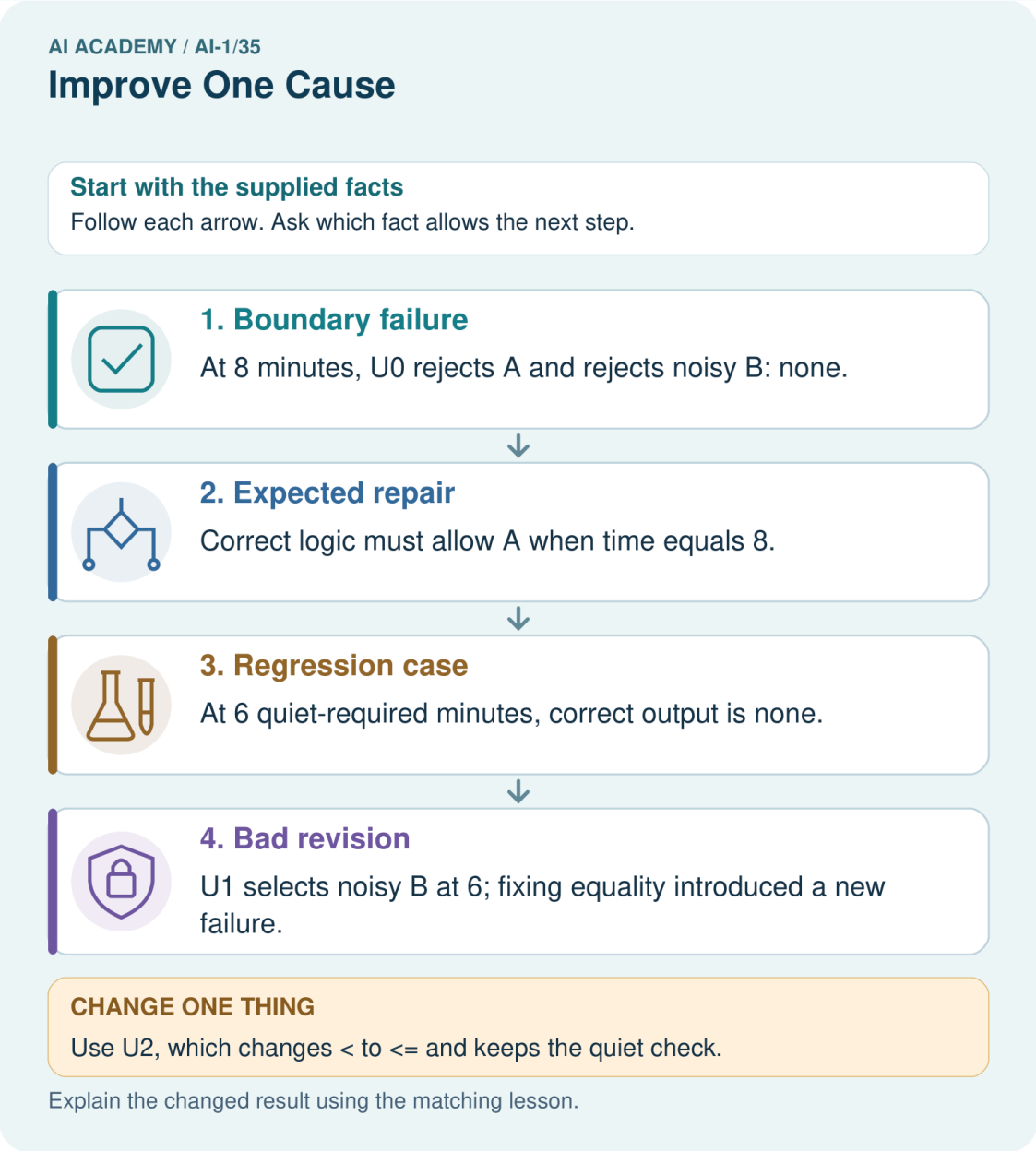
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Improve One Cause

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: Plan and operate the S1 **repair** and **regression** set described in the lesson, using the appropriate catalogue version for each case. Record expected and observed results separately with reasons and traces.

Summarise full passes over all attempted cases, list unrun cases separately and identify any remaining failures. Do not replace observed mistakes with worked answers.

Expected reasoning and fading help

Teacher check: Under B version 1 the listed requests yield B1, B2, no match, no match, B1, B2, no match, B1, then four clarifications for 31, blank minutes, 10.5 and Maybe. B version 2 with 12 Yes yields B3.

B version 3 yields B4 for 15 No and no match for 15 Yes. There are 15 planned cases. Correct execution with correct reasons gives 15 of 15; accept honest different **observation**s with investigation.

Unrun cases are not passes and the attempted denominator must be stated.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for Improving and checking **regression**s. Proposed explanation: “One **repair**ed example proves improvement everywhere”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your **repair** if needed.

Expected correction: Check the revealing case and previously successful behaviour under recorded conditions.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Explain why “our **repair**ed version now picks something” is not enough to show success at 6.

Any chosen activity must be quiet and fit the time. B is noisy, so picking it violates the contract; none is the correct **output**.

Changed-case reasoning: At 8 return A; at 6 return none. Equality is **repair**ed while B remains ineligible because it is noisy.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: The team changes lighting, labels and rules together; errors fall from four to one.

Write the question to explain: Can it identify the cause of improvement?

Reveal after **prediction**s: No. Several factors changed. Compare one planned change at a time with controlled conditions.

Use the earlier changed question as optional later practice. Reserve Audit a different proposed repair for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

For 9 Yes, required no match; S0 met it but T1 newly returns nonquiet B2 because quiet was removed, so a **regression**. For 8 No, required B2; S0 already excluded equality and returned no match, so an existing **defect**.

Keep saved versions, **change record**s, exact **input**s and catalogues, expected and observed reasons, traces and results for the remaining routine, validation and variation cases.

Exit: one **repair**ed case does not check earlier behaviour; deleting the old trace destroys **evidence** of the **defect**. Preserve the failure and rerun the wider development set with a new version.

Fixing the candidate before reveal prevents tailoring that candidate to the reserved cases.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the inclusive time requirement and S0 equality failure.
7 to 20 minutes	Read the repair diagram and complete W1 with a specific comparison change.
20 to 33 minutes	Trace S0, T1 and S1; complete W3 and distinguish old defects from new regressions.
33 to 45 minutes	Begin W2 partner repair checks using saved versions and prepared expectations.
Session 2 start to 8 minutes	Continue W2 with validation and catalogue variations; report attempted and unrun cases honestly.
8 to 23 minutes	Complete the guided explanation and organise the candidate evidence folder.
23 to 38 minutes	Collect W4 independently before support; check first-fit and equality reasoning.
38 to 45 minutes	Complete W5 and the exit, record unresolved issues and fix a candidate for final evaluation.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

For the supplied repair cycle, S1 must return B1,B2,no match,no match on the original R set; B1 and B2 on equality cases 10 Yes and 8 No; no match and B1 on valid endpoints 0 Yes and 30 Yes; clarification for 31, blank, fractional minutes and Maybe quiet. B version 2 gives B3 for 12 Yes; B version 3 gives B4 for 15 No and no match for 15 Yes. Collect actual runs rather than transcribing these predictions. T1's B2 at 9 Yes is a new quiet regression; S0's failure at 8 No was an existing equality defect. If the saved learner version already passes, accept no change with retained evidence. Reserve Audit a different proposed repair for an independent first attempt.

Run and record

1. Complete the task: Revise one rule or **input** and rerun the same tests.
2. Keep a normal case and an edge case clearly separated.
3. Record what happened—even when it does not match your **prediction**.
4. Explain the result, one limitation, and the next useful test.

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

One **repair**ed example proves improvement everywhere: Check the revealing case and previously successful behaviour under recorded conditions.

Any failure after editing is a **regression**: Show that the same behaviour was correct before the change; otherwise it may be an existing **defect**.

A good project must contain a **repair**: An unchanged correct version can be justified with honest **evidence**.

Choose support from the response

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Predict first, test honestly, and explain what the **evidence** means.

State your **prediction** before the result is known.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Audit a different proposed repair for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

CLAIM TO **REPAIR** An improvement proves every change was useful.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A seen final case becomes unseen after a new version label: Once it guides revision, it is development **evidence** for that revision.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: Equipment catalogue E lists E1 first with capacity 4 and E2 second with capacity 8. Valid whole group sizes are 1 to 10.

Choose the first capacity at least the group size, or no match; invalid **input** needs clarification. Old U0 uses strictly greater capacity and stops at the first fit.

Proposed U1 includes equality but keeps searching and returns the last fit. Independently compare U0, U1 and the requirement for groups 8 and 3. Identify the **repair** and **regression**, propose U2 and give one validation check.

No booking occurs.

For 8, U0 returns no match, U1 returns E2 and the requirement is E2: equality is **repair**ed. For 3, U0 returns E1 correctly but U1 returns E2, violating first-fit order: a **regression**.

U2 includes equality and stops immediately at the first fit, giving E2 for 8 and E1 for 3. Preserve both old traces and rerun both cases plus wider tests.

Group 0 or 11, a blank or a nonwhole size needs clarification before searching.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to **repair** it. Have the learner make that **repair** now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Write the change record

Use S0's failure on 10 Yes with catalogue B version 1. Record the expected and incorrect **output**s, the faulty time comparison, the exact S1 correction and the other conditions that must remain.

Explain what old **evidence** to preserve.

Expected B1 with equality and quiet satisfied; S0 incorrectly returns no match because its time condition is strictly less than. S1 uses no more than, including equality.

Preserve validation, quiet, first-match order, advancement, stopping and reset. Save S0, its failing trace and exact request/catalogue labels beside the new S1 **change record** and rerun.

W2 Check the repair and earlier behaviour

Plan and operate the S1 **repair** and **regression** set described in the lesson, using the appropriate catalogue version for each case. Record expected and observed results separately with reasons and traces.

Summarise full passes over all attempted cases, list unrun cases separately and identify any remaining failures. Do not replace observed mistakes with worked answers.

Under B version 1 the listed requests yield B1, B2, no match, no match, B1, B2, no match, B1, then four clarifications for 31, blank minutes, 10.5 and Maybe. B version 2 with 12 Yes yields B3.

B version 3 yields B4 for 15 No and no match for 15 Yes. There are 15 planned cases. Correct execution with correct reasons gives 15 of 15; accept honest different **observation**s with investigation.

Unrun cases are not passes and the attempted denominator must be stated.

W3 Reject a repair that breaks quiet requests

T1 accepts equal times but omits the quiet check. Explain why it passes 10 Yes yet fails 9 Yes. Use S0's earlier correct result for 9 Yes to justify the word **regression**.

What should change in T1, and which two cases should be rerun first?

At 10 Yes, first B1 fits and happens to be quiet, so the omission is hidden. At 9 Yes, T1 selects B2 on time despite its nonquiet label; required **output** is no match.

S0 correctly gave no match on that request, so the new failure is a **regression**. Restore the quiet condition combined with inclusive time; rerun 10 Yes for the **repair** and 9 Yes for the **regression**, then the wider set.

W5 Protect the meaning of final evaluation

After seeing a reserved final case fail, a team edits its candidate, reruns that same case and calls the result "a successful unseen test of the new version". Explain what is wrong and what should be recorded instead.

State what must be saved before an evaluation begins.

The case informed the **repair** and is now seen development **evidence** for the new version. Preserve the original candidate and its final result, label the revision and describe the repeated case as a **repair** check.

A fresh independent evaluation needs new reserved cases for the fixed revision. Save complete instructions, catalogue, requirements, version label, change history, development results and known limitations before evaluation.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

No. Several factors changed. Compare one planned change at a time with controlled conditions.

Additional transfer practice question

Dim photos fail. You change the lighting and all decision rules together; performance improves. Can you tell which change helped? Design a clearer comparison.

Reasoned practice answer

No. Change one factor at a time and compare the same test cases. Better lighting alone versus the original conditions would help isolate its effect.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “An improvement proves every change was useful.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

B has fewer recorded errors under changed conditions; no single change is proven responsible. Compare the same fixed version and comparable cases with only lighting changed, repeat where useful and keep labels constant.

Reserve fresh tests for the selected final version.

Explanations for the additional workbook examples

W1 Write the change record

W2 Check the repair and earlier behaviour

W3 Reject a repair that breaks quiet requests

W4 Audit a different proposed repair

Equipment catalogue E lists E1 first with capacity 4 and E2 second with capacity 8. Valid whole group sizes are 1 to 10. Choose the first capacity at least the group size, or no match; invalid **input** needs clarification.

Old U0 uses strictly greater capacity and stops at the first fit. Proposed U1 includes equality but keeps searching and returns the last fit. Independently compare U0, U1 and the requirement for groups 8 and 3.

Identify the **repair** and **regression**, propose U2 and give one validation check. No booking occurs.

For 8, U0 returns no match, U1 returns E2 and the requirement is E2: equality is **repair**ed. For 3, U0 returns E1 correctly but U1 returns E2, violating first-fit order: a **regression**.

U2 includes equality and stops immediately at the first fit, giving E2 for 8 and E1 for 3. Preserve both old traces and rerun both cases plus wider tests.

Group 0 or 11, a blank or a nonwhole size needs clarification before searching.

W5 Protect the meaning of final evaluation

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

If the sorter has an **evidence**d defect, save a new version with one **evidence**-linked change and a before/after test table; otherwise retain the current tested version and state that no repair was needed.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Maintain sorter **regression** tests and compare complete version traces, retaining new failures rather than hiding them.

Support while preserving the reasoning goal

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Use separate cards for time or capacity, quiet, and first-match stopping so pupils can point to the changed condition. Provide change-record headings without supplying the diagnosis.

Partners may divide execution and recording, then swap; collect the independent audit before hints. Extension: explain why a higher total can still conceal a serious **regression** and why individual failures matter.

W4 must identify U0 no match and U1 E2 for group 8, then U0 E1 and U1 E2 for group 3, with equality **repair**ed but first-match stopping broken. U2 needs both inclusive capacity and immediate first-fit stopping.

After support, retry with F1 capacity 5 then F2 capacity 9, same **input** range: group 9 exposes the equality **defect** and group 2 exposes the last-fit **regression**. Require fresh reasoning and one valid clarification test.

Plan the delayed check

Suggested return points in this level: ai-1/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 36: Expo and Defense. Retrieve today's **evidence** and identify what carries forward and what is new.

Week 36 evaluates a **fixed candidate** with reserved cases and asks pupils to defend their project using results, failures, a baseline and explicit limitations.

▼ **Earlier wording, alternatives and picture transcripts**

Connect and extend learning

Helpful Classroom Sorter

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: **Repair** a located **defect** without changing the agreed requirements

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: Plan and operate the S1 **repair** and **regression** set described in the lesson, using the appropriate catalogue version for each case.

Record expected and observed results separately with reasons and traces. Summarise full passes over all attempted cases, list unrun cases separately and identify any remaining failures.

Do not replace observed mistakes with worked answers.

Reasoned practice answer: Under B version 1 the listed requests yield B1, B2, no match, no match, B1, B2, no match, B1, then four clarifications for 31, blank minutes, 10.5 and Maybe. B version 2 with 12 Yes yields B3.

B version 3 yields B4 for 15 No and no match for 15 Yes. There are 15 planned cases. Correct execution with correct reasons gives 15 of 15; accept honest different **observation**s with investigation.

Unrun cases are not passes and the attempted denominator must be stated.

Check every learner before discussion: Use the worked case for Improving and checking **regression**s. Proposed explanation: “One **repair**ed example proves improvement everywhere”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your **repair** if needed.

Misconception **repair**: Check the revealing case and previously successful behaviour under recorded conditions.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Use U2, which changes < to ≤ and keeps the quiet check. Require a **prediction** and a reason before comparing. Expected result: At 8 return A; at 6 return none.

Reason: Equality is **repair**ed while B remains ineligible because it is noisy.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ Reference: feedback and records

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ Earlier wording, alternatives and picture transcripts

This week has no code block. Use this routine when a later lesson includes an executable lab.

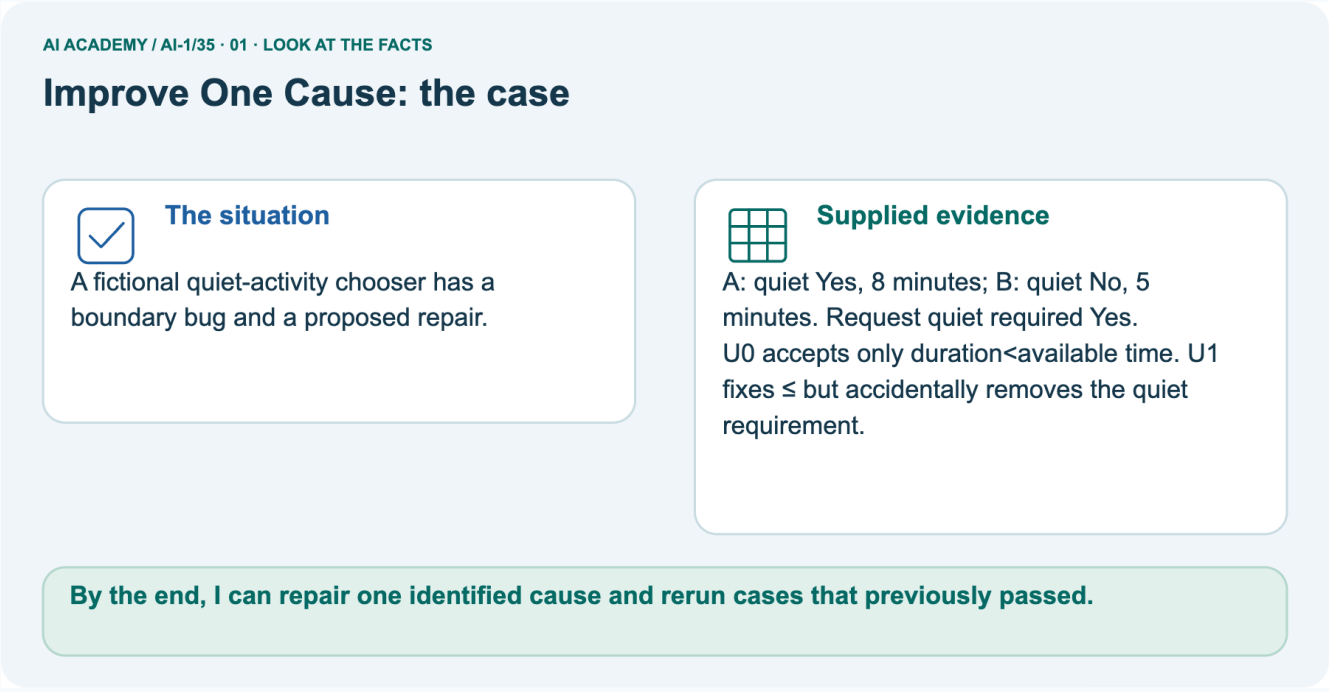
Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

▼ Reference: teach with the six pictures

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Improve One Cause: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

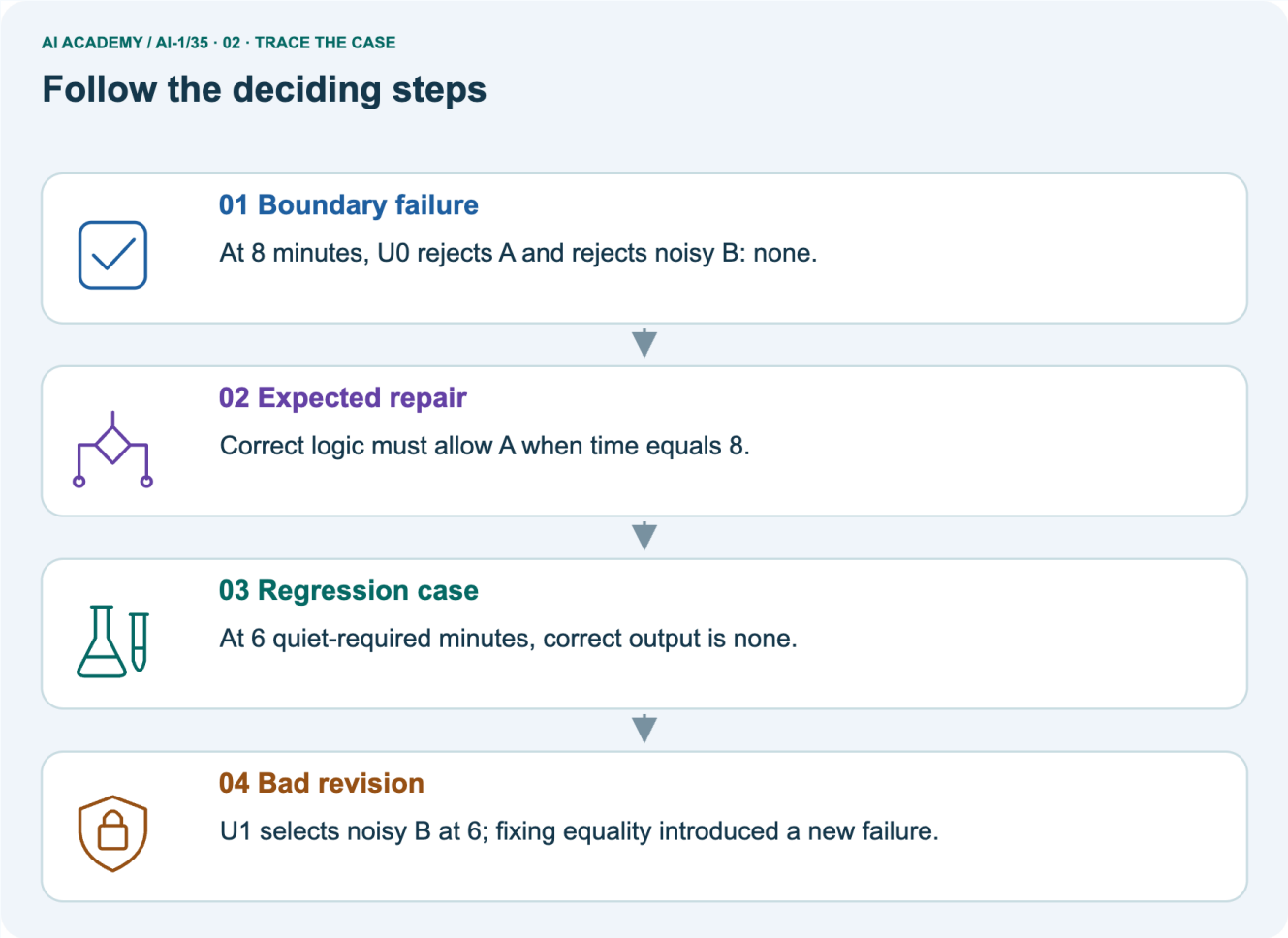
A fictional quiet-activity chooser has a boundary bug and a proposed **repair**. Supplied facts: A: quiet Yes, 8 minutes; B: quiet No, 5 minutes. Request quiet required Yes. U0 accepts only duration<available time.

U1 fixes ≤ but accidentally removes the quiet requirement.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



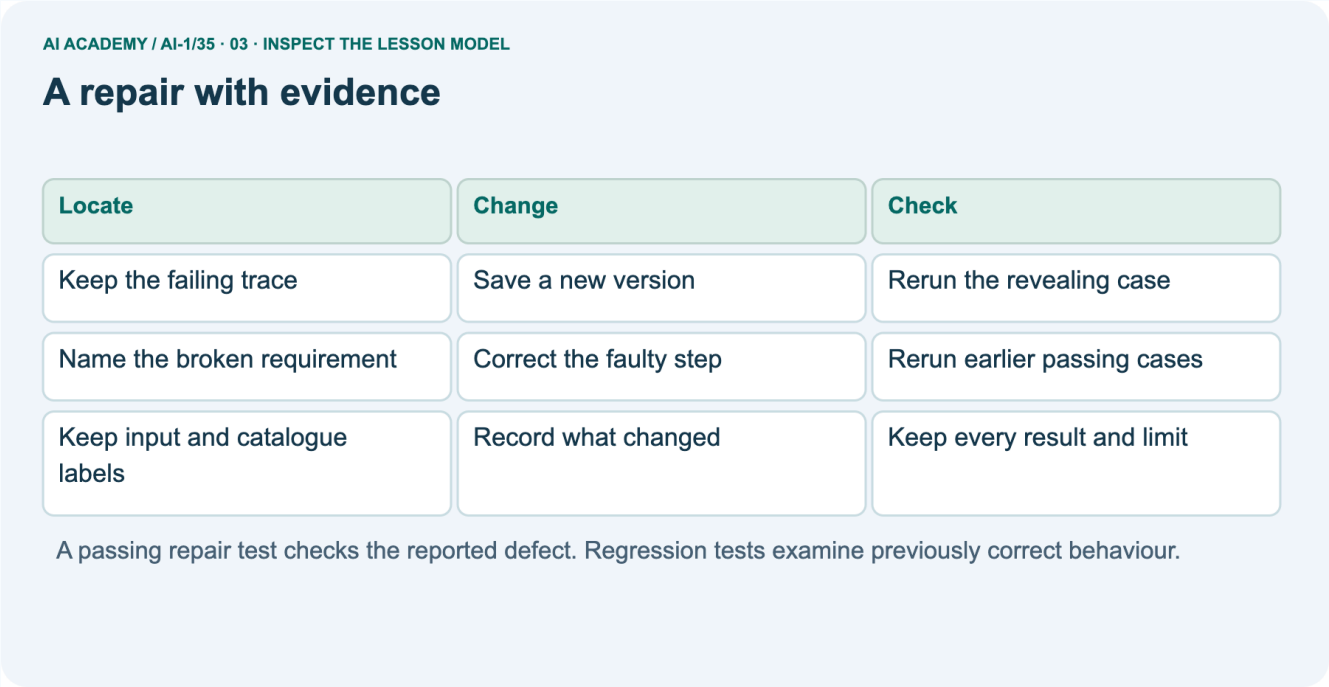
Follow the labelled picture with the matching explanation. Open the vector diagram

Boundary failure: At 8 minutes, U0 rejects A and rejects noisy B: none. → Expected **repair**: Correct logic must allow A when time equals 8. → **Regression** case: At 6 quiet-required minutes, correct **output** is none. → Bad revision: U1 selects noisy B at 6; fixing equality introduced a new failure.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. A repair with evidence



Follow the labelled picture with the matching explanation. Open the vector diagram

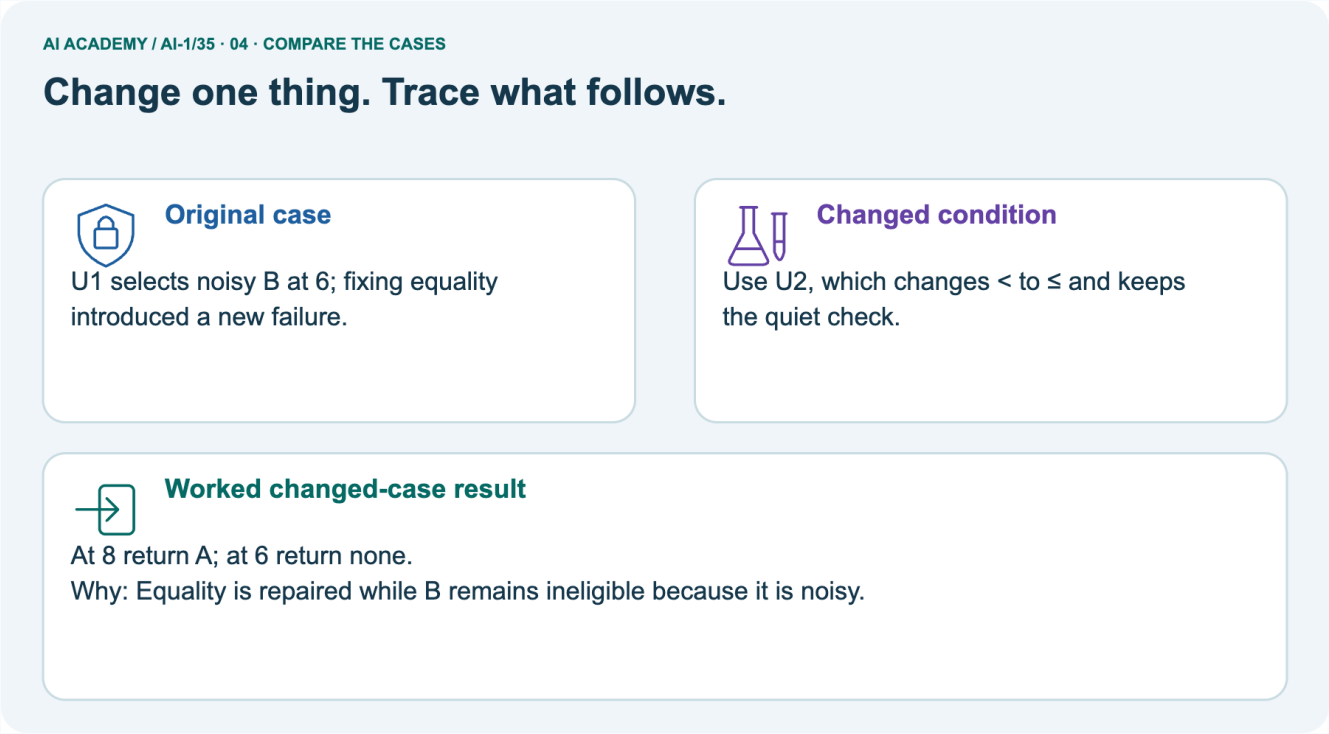
A **repair** with **evidence**. Keep the failing trace | Save a new version | Rerun the revealing case. Name the broken requirement | Correct the faulty step | Rerun earlier passing cases.

Keep **input** and catalogue labels | Record what changed | Keep every result and limit

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: U1 selects noisy B at 6; fixing equality introduced a new failure. Changed condition: Use U2, which changes < to ≤ and keeps the quiet check. Expected result: At 8 return A; at 6 return none.

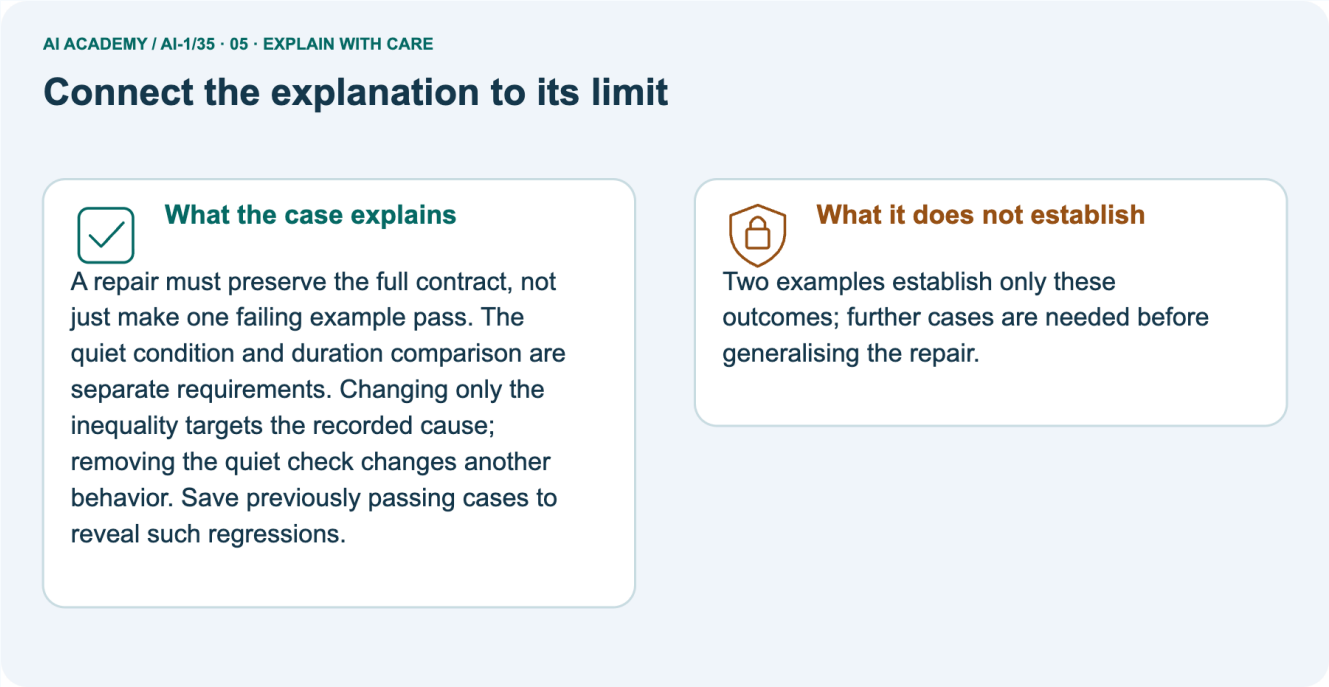
Reason: Equality is **repair**ed while B remains ineligible because it is noisy.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Equality is **repair**ed while B remains ineligible because it is noisy.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **repair** must preserve the full contract, not just make one failing example pass. The quiet condition and duration comparison are separate requirements.

Changing only the inequality targets the recorded cause; removing the quiet check changes another behavior. Save previously passing cases to reveal such **regression**s.

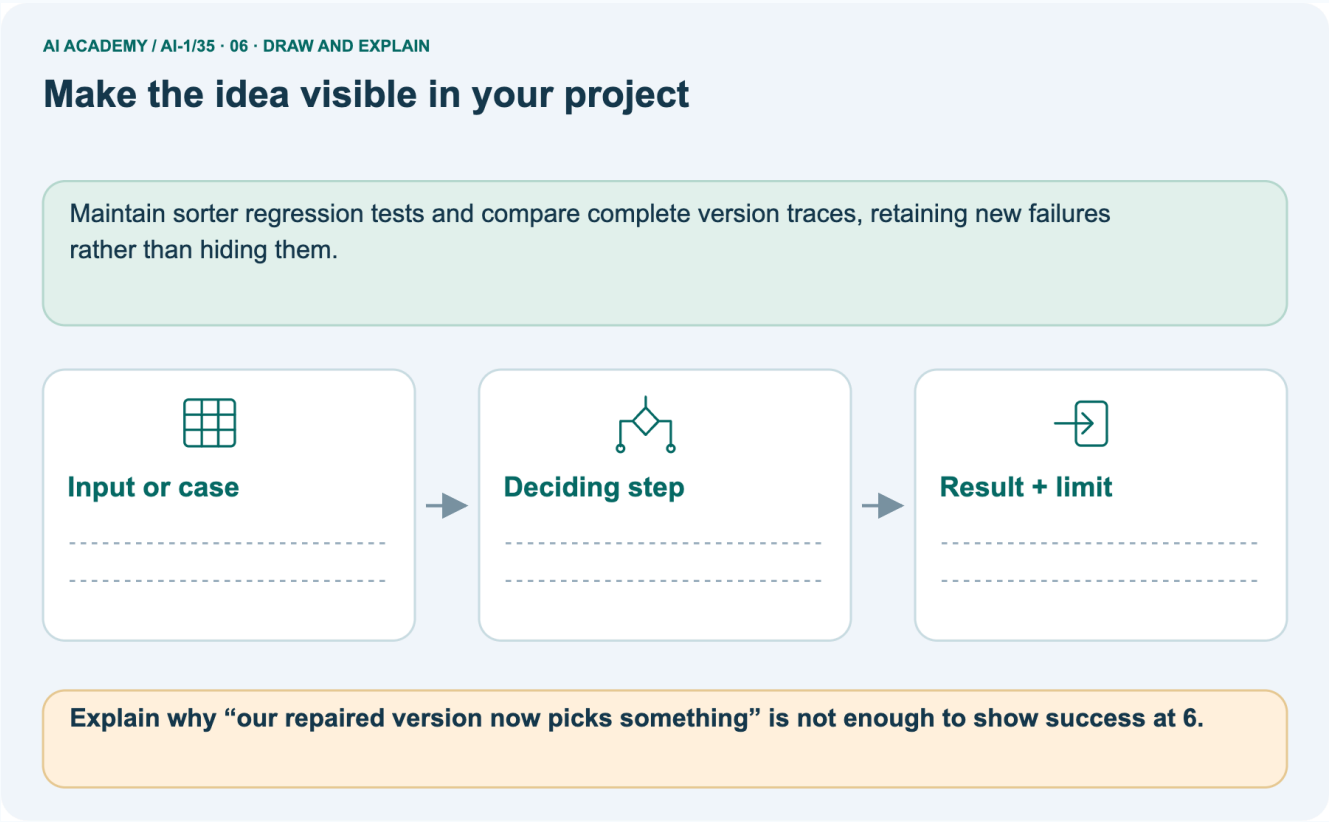
Limit: Two examples establish only these outcomes; further cases are needed before generalising the **repair**.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Two examples establish only these outcomes; further cases are needed before generalising the **repair**.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain sorter **regression** tests and compare complete version traces, retaining new failures rather than hiding them. Draw your **input**, deciding step, result and limit.

Supported question: Explain why “our **repair**ed version now picks something” is not enough to show success at 6.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Any chosen activity must be quiet and fit the time. B is noisy, so picking it violates the contract; none is the correct **output**.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Improve One Cause: the case

02. Follow the deciding steps

03. A repair with evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.